

Topic	Description
Module	Non Linear Electronics
Module Code	MECH-M-1-EDY-NLE-VO
Semester	WS 2025
Lecturer	Daniel T. McGuiness, Ph.D
ECTS	3
SWS	2
Lecture Type	ILV
Coursework Name	Individual Assignment
Work	Individual
Suggested Private Study	10 hours
Submission Format	Submission via SAKAI
Submission Deadline	25.01.2026 23:59
Late Submission	Not accepted
Resubmitting Opportunity	No re submission opportunity

No lecture time is exclusively devoted to the aforementioned assignment.

A portion of the mark for every assignment will be, where applicable, based on style. Style, in this context, refers to organisation, flow, sentence and paragraph structure, typographical accuracy, grammar, spelling, clarity of expression and use of correct IEEE style for citations and references. Students will find *The Elements of Style (3rd ed.)* (1979) by Strunk & White, published by Macmillan, useful with an alternative recommendation being *Economist Style Guide (12th ed.)* by Ann Wroe.

Submission Format and Requirements with L^AT_EX

Before submitting your work for assessment please follow the following requirements.

1. **The work must be done using L^AT_EX and no other format will be accepted.** You are allowed to use whatever software you wish to write (i.e., Overleaf, T_EXstudio, LyX, ...). The work must also conform to the **MCI Documentation guidelines** (which is in SAKAI). A template is provided to you [here](#) which was written by the author of this document.

If you are to use **mcidoc** template, set the **document-state** as **Report**.

To learn L^AT_EX, the author has tutorials which can be accessed [here](#).

Please treat this assignment as a great practice to learn L^AT_EX as you will need it when you are writing your thesis.

2. The report (and all its content) **must** be submitted as a **.pdf** along with all the necessary files (**.tex**, **.sty**) to compile it. Submit one (1) **.zip** file containing everything with the following name pattern.

[STUDENT-ID]_[STUDENT-NAME]_[STUDENT-SURNAME].zip

For example, a student named **Jason Brigham** with a student ID **2710704051** would send their submission as:

2710704051_Jason_Brigham.zip

and their submission would be in arranged as the following directory:

2710704051_Jason_Brigham/

```
├── tex/
│   ├── 2710704051_Jason_Brigham.tex, *.pdf, mcidoc.cls, custom.sty, ...
│   └── figures/
│       ├── .jpg, .png, ...
└── HW_Content/
    ├── src/, include/, .cpp, .bash, .py, ...
```

where **...** are any additional requirement (i.e., **.bib**, **.glo**) you may need for T_EX and the **HW_Content** is any content which is relevant to the Homework (i.e., if your HW is about python, this is where all the python scripts would be kept, or where your ROS 2 source files would be, or where your C++ codes would be for a project.)

3. The work must be submitted to SAKAI and **no link to personal repo will be accepted.**

To make sure your submission conform to given requirements, please test out zipping/un-zipping and running any code before submission. Any deviations from the requirements will incur penalties to the final grade.

Question	Maximum Point	Received Point
Transfer Function of an Op-Amp Circuit	15	
Small Signal Analysis of a Diode	15	
Ideal Diode Path	10	
MOSFET as a Linear Resistor	20	
Analysis of a npn Transistor	40	
Sum	100	

[Q1] Transfer Function of an Op-Amp Circuit _____ 15

- a. Find the transfer function of the following circuit. (5)

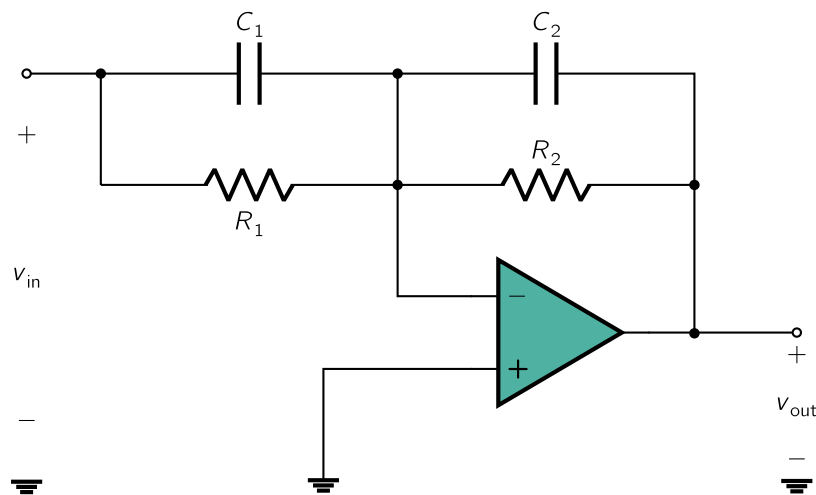


Figure 1: The circuit for the question "Transfer Function of an Op-Amp Circuit".

- b. What is the condition for which the output is independent of frequency? Using a programming language of choice (i.e., Python, SPICE, MATLAB) draw the Bode magnitude plots (in rad s^{-1}) for the following three (3) cases: (10)
- $C_2 = 0.1$ $C_1 = 0.1 \mu\text{F}$ $R_2 = 10R_1 = 100 \text{ k}\Omega$
 - R_2 is raised to $1 \text{ M}\Omega$,
 - R_2 is lowered to $10 \text{ k}\Omega$.

In your report include the code as a snippet (you can use `listings` or `minted` for this) and make sure you put the file (`.py`, `.m`, `.cir`) within your `HW_Content/src` as required by the **Submission Format and Requirements with L^AT_EX**

[Q2] Small Signal Analysis of a Diode 15

A junction diode for which $n = 2$ operates in a particular circuit with a current which varies over the range 0,1 mA to 10 mA. What is the diode incremental resistance at the extreme values of the current?

If you were asked to state an “average” resistance, which would be the best, arithmetic mean $(r_1 + r_2)/2$ or geometric mean $(r_1 r_2)^{1/2}$? Please calculate both, as well as the resistances at 5,05 mA and at 1 mA (The mean currents of each kind). Please evaluate and discuss your results.

[Q3] Ideal Diode Path 10

Find the currents, I_1, I_2, I_3, I_4 in each of the diodes D_1, D_2, D_3, D_4 of the circuit shown in Fig. 2. What would be the voltage measured at v_{out} . Assume your diodes to be ideal.

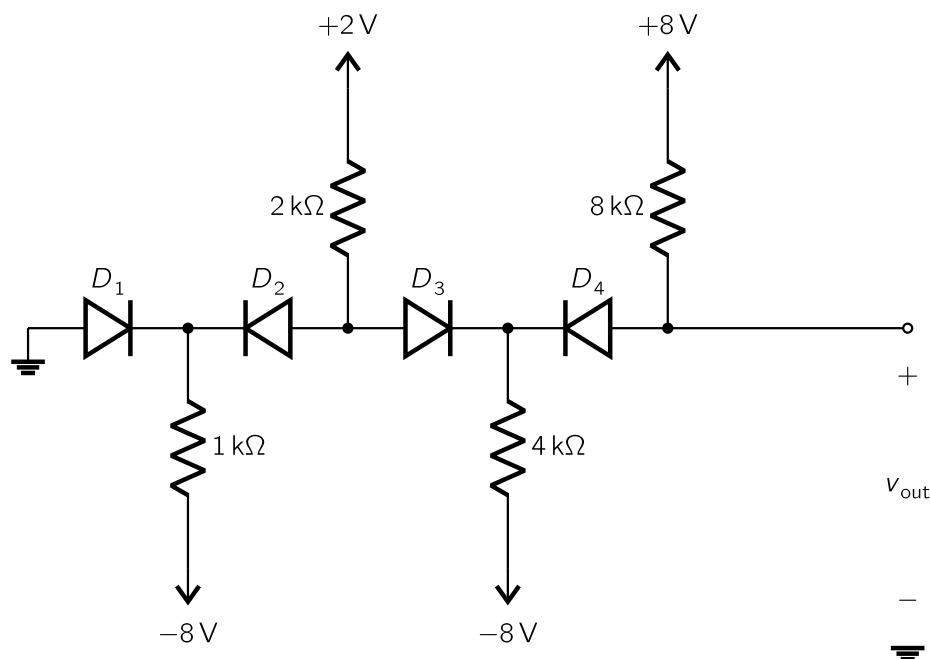


Figure 2: The circuit for the question “Ideal Diode Path”.

[Q4] MOSFET as a Linear Resistor 20

An n -channel enhancement MOSFET having $\mu_n C_{ox} = 20 \mu A V^{-2}$, $W = 20 \mu m$, $L = 2 \mu m$, and $V_t = +1 V$ is to be used for small signals as a linear resistor in range $1 k\Omega$ to $1 M\Omega$. What corresponding range of values of v_{GS} are required?

What are the corresponding ranges of operating current (i_D) and voltage (v_{DS}) for which the resistance provided is within 10% of its desired value?

[Q5] Analysis of a npn Transistor 40

A particular *npn* transistor has an emitter area of $20\text{ }\mu\text{m} \times 20\text{ }\mu\text{m}$. The doping concentrations are: $N_D = 1 \cdot 10^{19}\text{ cm}^{-3}$ in the emitter, $N_A = 1 \cdot 10^{17}\text{ cm}^{-3}$ in the base, and $N_D = 1 \cdot 10^{15}\text{ cm}^{-3}$ in the collector.

The transistor is assumed to operate at $T = 300\text{ K}$, where $n_i = 1,5 \cdot 10^{10}\text{ cm}^{-3}$. For electrons diffusing in the base: $L_n = 19\text{ }\mu\text{m}$ and $D_n = 21,3\text{ cm}^2\text{ s}^{-1}$. For holes diffusing in the emitter, $L_p = 0,6\text{ }\mu\text{m}$ and $D_p = 1,7\text{ cm}^2\text{ s}^{-1}$ and $q = 1,6 \cdot 10^{-19}\text{ C}$.

For base widths $W = 1\text{ }\mu\text{m}$ and $0,1\text{ }\mu\text{m}$, and $v_{BE} = 700\text{ mV}$, please calculate:

$$n_{p0}, n_p(0), I_n, I_s, i_c, \beta, \alpha$$

Report You are to write a report on your individual assignment which should explain the calculation used in solving the given questions. A template structure for this assignment could be as follows. Per each question:

- An introduction to the question, and the method used to tackle.
- Explanation (i.e, conceptual and mathematical) of the method(s) used in the tackling of the problem. Calculations should be explicit and detailed with proper units and calculations clearly written.
- If any code is used to do calculation/drawing a requested figure, the code must be included within the report as a snippet and be explained and in addition the code itself must be within your **HW_Content/src** as required by the **Submission Format and Requirements with L^AT_EX**

Before submission, please read **Submission Format and Requirements with L^AT_EX**.